

① The PDE: We are concerned w/ tracking evolution of:
 $p(t, x)$:= phenotypic density function, which is
 the density of cells in phenotypic state $x \in \Omega \subset \mathbb{R}$
 at time $t \geq 0$.

$$\Omega := [l, L], \text{ where } 0 < l < L.$$

Note: x is one continuous phenotype variable

Ex: x = drug resistance level or x = expression of some marker gene ...

$$x \in [0, 1]$$

↑ very sensitive to drug ↓ very resistant

We then can define

$P(t)$:= population size at time t , where

$$P(t) = \int_{\Omega} p(t, x) dx$$

$x \in \Omega$ (phenotype space)

(global quantity)
is $P(t) = \int_{\Omega} \dots$

The phenotype-structured PDE we are considering is a nonlocal reaction-diffusion-advection equation:

$$\partial_t p(t, x) - \beta \partial_{xx}^2 p(t, x) + \partial_x [V(t, x) p(t, x)] = \left(f(t, x) - \frac{P(t)}{K} \right) p(t, x)$$

which has no-flux boundary condition (cells cannot leave admissible phenotype interval)

$$\beta \partial_x p - V(t, x) p = 0 \quad \text{for } x \in \partial\Omega$$

and initial condition $\text{nonnegative cell density}$ ($\partial\Omega$ = boundary of Ω)

$$p(0, x) = p^0(x) \geq 0 \quad \text{with} \quad \int_{\Omega} p^0(x) dx > 0 \quad \leftarrow \text{positive total population}$$

Terms of PDE:

② $\beta \partial_{xx}^2 p(t, x)$:= models spontaneous phenotypic changes as diffusive flux.

③ $\partial_x [V(t,x) p(t,x)] :=$ models environment driven phenotypic changes w/ advection term w/ velocity $V(t,x)$.

③ $f(t,x) p(t,x) :=$ phenotype - dependent growth. w/ rate $f(t,x)$

④ $-\frac{P(t)}{K} p(t,x) :=$ death due to competition for space which is dependant on population size $P(t)$. $K =$ carry-cap.

Note: assume $f(t, \cdot) \in C^0(\mathcal{X})$, $V(t, \cdot) \in C^0(\mathcal{X})$ $\forall t$ $f_m \leq f(\cdot, x) \leq f_M$
 $V_m \leq V(\cdot, x) \leq V_M$

2. Motivation / Problem.

PDE models tracking evolution of phenotypic density distribution of cell population have high analytical and computational costs of solving compared to ODEs.

① PDE solvers often implemented on a problem specific basis.

② Parameter estimation requires many simulations.

③ Identifiability / sensitivity analysis tools more developed for ODEs.

④ Data may only measure total population, mean phenotype, variance, not the full distribution.

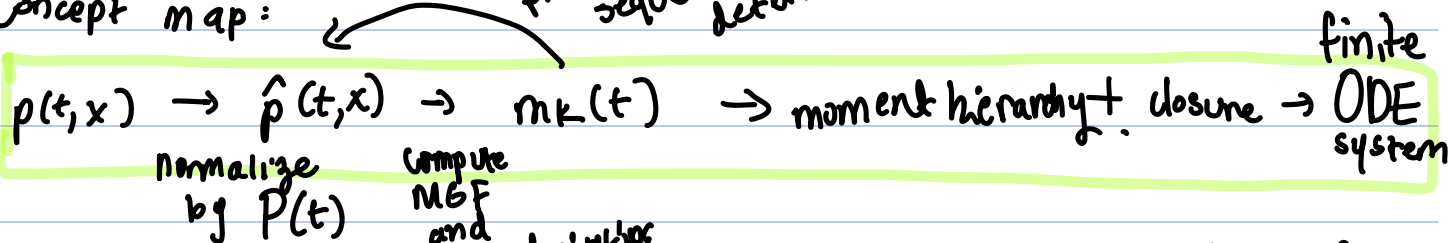
Can we keep the interpretability of the PDE but get an ODE system?

3. Main Idea: We will recast the phenotype density function as a probability distribution, derive equations (ODEs) for moments of this distribution, then close the infinite hierarchy of moments to obtain a finite

ode system.

* Concept map:

Stone-Weierstrass thm
 2. Has diff moment problem says the sequence of moment determine $p(x)$



Goal: replace $p(t, x)$ with a few interpretable quantities, total pop, mean phenotype, variance.

4. Reduction of PDE to $P(t)$ + moments.

(a) Phenotype density function $\hat{p}(t, x)$ scaled by total population size gives us a probability density function over phenotype space, $\hat{p}(t, x)$ where

$$\hat{p}(t, x) = \frac{p(t, x)}{P(t)}$$

So we have: $\int_{\Omega} \hat{p}(t, x) dx = 1.$

Answers this in what we are reducing PDE to.

We note that $p(t, x)$ contains two pieces of information:

1. How many cells there are total

2. How those cells are distributed across phenotypes

1. corresponds to the first ODE we derive for $P(t)$

2. corresponds to the moment equations (moments summarize shape / comp of $\hat{p}$)

With this note, we will first derive the ODE for $P(t)$.

Since $P(t) = \int_{\Omega} p(t, x) dx,$

$$\frac{d}{dt}(P(t)) = \frac{d}{dt} \int_{\Omega} p(t, x) dx$$

$$= \int_{\Omega} \partial_t p(t, x) dx$$

← Assume enough regularity to do this

substitution
w/ PDE

$$= \int_{\Omega} \left[\beta \partial_{xx}^2 p(t, x) - \partial_x [V(t, x) p(t, x)] + \left(f(t, x) - \frac{P(t)}{K} \right) p(t, x) \right] dx$$

↓ split integral

$$= \int_{\Omega} \left[\beta \partial_{xx}^2 p(t, x) - \partial_x [V(t, x) p(t, x)] \right] dx + \int_{\Omega} \left(f(t, x) - \frac{P(t)}{K} \right) p(t, x) dx$$

$$= \left[\beta \partial_x p(t, x) - V(t, x) p(t, x) \right] \Big|_{\partial \Omega} + \int_{\Omega} \left(f(t, x) - \frac{P(t)}{K} \right) p(t, x) dx$$

↓ from b.c.

$$= 0 + \int_{\Omega} f(t, x) p(t, x) dx - \frac{P(t)}{K} \underbrace{\int_{\Omega} p(t, x) dx}_{P(t)}$$

$$\boxed{\frac{dP}{dt} = \int_{\Omega} f(t, x) p(t, x) dx - \frac{P^2(t)}{K}}$$

↪ This eqn. depends explicitly on $p(t, x)$ want to recast it as a probability distribution, so we can ... eventually ... study moments of this distribution

$$So: \quad \frac{dP}{dt} = P(t) \int_{\Omega} f(t, x) \hat{p}(t, x) dx - \frac{P^2(t)}{K}$$

$$\boxed{\frac{dP}{dt} = \left(\int_{\Omega} f(t, x) \hat{p}(t, x) dx - \frac{P(t)}{K} \right) P(t)}$$

i.e. from $P(t) = \int p(t, x) dx$
 $P(0) = \int p(0, x) dx$

↪ $P(0) > 0 \Rightarrow P(t) > 0 \quad \forall t$
 Pf: from f being bounded & using Gronwall's inequality.

Tool : Moment generating function.. + moments of $\hat{p}(t, x)$

The moment generating function is defined as

$$M(s, t) = \int_{\Omega} e^{sx} \hat{p}(t, x) dx$$

We define the k th moment ^{$m_k(t)$} of the distribution $\hat{p}(t, x)$ as

$$m_k(t) = \partial_s^k M(s, t) \big|_{s=0} \quad \text{for } k \geq 0.$$

i.e. $m_k(0) = \partial_s^k (M(s, 0)) \big|_{s=0}$

Substituting $s=0$ in $M(s, t)$, we see that

$$m_0(t) = \int_{\Omega} \hat{p}(t, x) dx = 1$$

Note: Since Ω is compact, the Stone-Weierstrass thm. & solution Hausdorff Moment problem guarantee the sequence of moments $\{m_k(t)\}_{k=1}^{\infty}$ uniquely determine distribution $\hat{p}(t, x)$.

Deriving moment equations (General Approach):

Given $M(s, t) = \int_{\Omega} e^{sx} \hat{p}(t, x) dx$ and $m_k(t) = \partial_s^k M(s, t) \big|_{s=0}$

we can say now: $\downarrow$ Again assuming int $\Leftrightarrow$ diff.

$$m_k(t) = \int_{\Omega} \partial_s^k e^{sx} \hat{p}(t, x) dx \big|_{s=0}$$

$$m_k(t) = \int_{\Omega} x^k \hat{p}(t, x) dx$$

We want to derive ODEs for $m_k(t)$ for $k \geq 1$.

We can say

$$m_k'(t) = \frac{d}{dt} \int_{\Omega} x^k \hat{p}(t, x) dx = \int_{\Omega} x^k \hat{p}_t(t, x) dx$$

Substitute PDE after normalization here, integrate by parts, and simplify. PDE for $p \rightarrow$ PDE for $\hat{p}$

Concrete Case: The first moment, $m_1'(t)$.

m_1 is mean phenotype — tells us how avg. phenotype evolves.

$$m_1'(t) = \int_{\Omega} x \hat{p}_t(t, x) dx$$

need an eqn. for $\hat{p}$ not p .

$$\begin{cases} p = P \hat{p} \\ p_t = P_t \hat{p} + P \hat{p}_t \\ p_x = P \hat{p}_x ; p_{xx} = P \hat{p}_{xx} \end{cases}$$

With these the PDE becomes

$$P_t \hat{p} + P \hat{p}_t - \beta P \hat{p}_{xx} + P \partial_x [V(t, x) \hat{p}] = \left(f(t, x) - \frac{P(t)}{K} \right) P \hat{p}$$

Divide both sides by P :

$$\frac{P_t}{P} \hat{p} + \hat{p}_t - \beta \hat{p}_{xx} + \partial_x [V \hat{p}] = \left(f - \frac{P}{K} \right) \hat{p}$$

$$\frac{P_t}{P} = \left(\int_{\Omega} f(t, x) \hat{p}(t, x) dx - \frac{P(t)}{K} \right)$$

$$S_0: \hat{p}_t - \beta \hat{p}_{xx} + \partial_x [V \hat{p}] = \left(f - \int_{\Omega} f \hat{p} dx \right) \hat{p}$$

Describes evolution of normalized phenotype distribution.

Plugging $\hat{p}_t$ into RHS of m_1' ,

$$m_1'(t) = \int_{\Omega} x \left(\beta \hat{p}_{xx} - \partial_x [V \hat{p}] + \left(f - \int_{\Omega} f \hat{p} dx \right) \hat{p} \right) dx$$

$$m_1'(t) = \underbrace{\int_{\Omega} x \partial_x (\beta \hat{p}_x - [V \hat{p}]) dx}_{\text{IBP}} + \int_{\Omega} x \left(f - \int_{\Omega} f \hat{p} dx \right) \hat{p} dx$$

$$\text{IBP } u=x \quad dv = \partial_x (\beta \hat{p}_x - [V \hat{p}]) dx$$

$$du = dx \quad v = \beta \hat{p}_x - V \hat{p}$$

$$m_1'(t) = \underbrace{x(\beta \hat{p}_x - V \hat{p}) \Big|_{\partial \Omega}}_{0 \text{ due to b.c.}} - \int_{\Omega} (\beta \hat{p}_x - V \hat{p}) dx + \int_{\Omega} x f \hat{p} dx - \int_{\Omega} f \hat{p} dx \underbrace{\int_{\Omega} x \hat{p} dx}_{m_1(t)}$$

$$= - \int_{\Omega} \beta \hat{p}_x dx + \int_{\Omega} V \hat{p} dx + \int_{\Omega} x f \hat{p} dx - m_1 \int_{\Omega} f \hat{p} dx$$

$$m_1'(t) = -\beta [\hat{p}]_{\partial \Omega} + \int_{\Omega} V \hat{p} dx + \int_{\Omega} x f \hat{p} dx - m_1 \int_{\Omega} f \hat{p} dx$$

Eqn. for m_1 . For higher moments, the same general process applies, but for the k th moment eqn.

$$\begin{aligned} \frac{d}{dt} m_k(t) &= \beta k(k-1) m_{k-2}(t) - \beta k \left[x^{k-1} \hat{p}(t, x) \right]_{\partial \Omega} + \dots \\ &\dots + k \int_{\Omega} x^{k-1} V(t, x) \hat{p}(t, x) dx + \int_{\Omega} x^k f(t, x) \hat{p}(t, x) dx - \dots \\ &\dots - m_k(t) \int_{\Omega} f(t, x) \hat{p}(t, x) dx \quad , k \geq 2. \end{aligned}$$

Note on boundary contribution assumption: The paper assumes that the boundary - density contribution is negligible, that is $[\hat{p}(t, x)]_{\partial \Omega} \approx 0$

Which is biologically motivated in that the phenotype distribution has very little mass near the phenotype boundaries.

So that now $m_1(t)$ can be written as

$$m_1'(t) = \int_{\Omega} V \hat{p} dx + \int_{\Omega} x f \hat{p} dx - m_1 \int_{\Omega} f \hat{p} dx$$

Moment eqns.

and

$$m_k'(t) = \beta k(k-1) m_{k-2}(t) + k \int_{\Omega} x^{k-1} V \hat{p} dx + \int_{\Omega} x^k f \hat{p} dx - m_k(t) \int_{\Omega} f \hat{p} dx \quad k \geq 2$$

These are the moment equations, but they still contain integrals involving $\hat{p}(t, x)$, so the system is not yet a closed moment system.

Further assumptions to get a closed moment system.

① Domain restriction

So far, we have equations for $P, m_k(t)$ for $k \geq 1$ involving

$\hat{p}$ so not closed. The first step to making this a

closed system is to consider a bounded phenotypic domain restricted to $\Omega = [0, 1]$ (note: before it was $[l, L]$... rescale can be done w/ $y = \frac{x-l}{L-l}$).

This restriction means $0 \leq x \leq 1 \Rightarrow 0 \leq m_k \leq 1$

This gives bounded moments & makes the Taylor/moment expansion easier to control.

(2) f, V Analytic
 Additionally, we consider $f(t, x)$ and $V(t, x)$ to be analytic in phenotype space. Therefore, we can Taylor expand f and V about $m_1(t)$.

$$f(t, x) = \sum_{n=0}^{\infty} f_n(t) (x - m_1(t))^n$$

→ natural center of phenotype distribution.

$$V(t, x) = \sum_{n=0}^{\infty} V_n(t) (x - m_1(t))^n, \text{ where}$$

$$f_n(t) = \frac{\partial_x^n f(t, x)}{n!} \Big|_{x=m_1(t)}, \quad V_n = \frac{\partial_x^n V(t, x)}{n!} \Big|_{x=m_1(t)}$$

(3) Integrals become moments.

Replace f and V in moment equations w/ Taylor expansion
 Consider eqn. derived first for pop. size

$$\frac{dP}{dt} = \left(\int_{\Omega} f(t, x) \hat{p}(t, x) dx - \frac{P(t)}{\kappa} \right) P(t)$$

problem b/c $\hat{p}$

With the Taylor expansion for f , we can write

$$\int_{\Omega} f(t, x) \hat{p}(t, x) dx = \sum_{n=0}^{\infty} f_n(t) \int_{\Omega} (x - m_1(t))^n \hat{p}(t, x) dx$$

paper rewrites w/ binomial th
 $\sum_{i=0}^n \binom{n}{i} x^{n-i} (-m_1(t))^i$

To see how this simplifies, consider second-order Taylor approx. for f :
 paper uses full center

$$f(t, x) \approx f_0(t) + f_1(t)(x - m_1(t)) + f_2(t)(x - m_1(t))^2$$

Then:

$$\int f(t, x) \hat{p}(t, x) dx = \int f_0(t) \hat{p}(t, x) dx + \int f_1(t)(x - m_1(t)) \hat{p}(t, x) dx + \int f_2(t)(x - m_1(t))^2 \hat{p}(t, x) dx$$

$$\begin{aligned}
 &= f_0(t) + f_1(t)m_1(t) - f_1(t)m_1(t) + f_2(t)(m_2(t) - 2m_1^2(t) + m_1^2(t)) \\
 &= f_0(t) + f_2(t)(m_2(t) - m_1^2(t))
 \end{aligned}$$

Therefore

$$\frac{dP}{dt} = \left(f_0(t) + f_2(t)(m_2(t) - m_1^2(t)) - \frac{P(t)}{K} \right) P(t)$$

→ We can do the same for $m_1(t), m_2(t), \dots, m_K(t)$

$$m_1'(t) = \int_{\mathcal{R}} V \hat{p} dx + \int_{\mathcal{R}} x f \hat{p} dx - m_1 \int_{\mathcal{R}} f \hat{p} dx$$

Taylor expand
& replace $\int_{\mathcal{R}} x^n \hat{p} dx = m_n(t)$

So that eventually,

$$P' = F_0(P, m_1, m_2, m_3, \dots)$$

$$m_1' = F_1(P, m_1, m_2, m_3, \dots)$$

$$m_2' = F_2(P, m_1, m_2, m_3, \dots)$$

So the Taylor expansion removes $\hat{p}$ but creates an infinite moment hierarchy. When using the full expansion, in addition to having already an infinite set of moment equations. So we need two approximations:

① Taylor series truncation for f & V

We choose some order say M to truncate the Taylor series

$$f(t, x) \approx \sum_{n=0}^M f_n(t) (x - m_1(t))^n$$

$$V(t, x) \approx \sum_{n=0}^M V_n(t) (x - m_1(t))^n$$

* Authors justify this by assuming f, V are smooth enough, domain is bounded, & higher order terms are small enough to neglect (assume $0 < m_1(t) \leq 1$, $(m_1(t))^i \rightarrow 0$ as $i \rightarrow \infty$)

② Moment closure: approximate higher moments

After Taylor truncation, the equations are written in terms of moments. They may still involve more moments than we want to solve for

$$\begin{aligned} m_1' &= F_1(m_1, m_2, m_3, \dots), \\ m_2' &= F_2(m_1, m_2, m_3, m_4, \dots). \end{aligned}$$

We can choose a moment order N , and assume phenotypic distribution $\hat{p}(t, x)$ is

Sufficiently well-characterized by a finite system of ODEs for 1st N moments.

→ Keep $m_1, \dots, m_N$.

→ If higher moments appear, approximate them:

$m_{N+1}, m_{N+2}, \dots$ in terms of $m_1, \dots, m_N$

* Two types of closure considered in paper

① Gaussian:

Set $m_k = m_1 m_{k-1} + (k-1)(m_2 - m_1^2) m_{k-2}$ for

$k = N+1, \dots, N+M$.

② Truncation:

Set $m_k = 0$ for $k = N+1, \dots, N+M$.

* Closure trustworthy when retained moments capture shape of distribution. Risky if dist. multimodal, skewed, or develops boundary mass.

Example 1: f & V are polynomials.

In this case we do not need to truncate Taylor series, but we do still need to close moment system.

$\mathcal{Q} = \mathbb{R}$ (can have since we do not need restrictions to truncate Taylor series)

* Point of example: Earlier papers assumed from start that the phenotype distribution was Gaussian. Authors show their general moment framework can recover same ODEs as special

case,

They take:

$$f(t, x) = a(t) - b(t)(x - X(t))^2$$

$$V(t, x) = V_0(t).$$

' $X(t)$: fittest phenotype.

• $a(t)$: maximum background fitness.

• $b(t)$: strength of selection.

• $V_0(t)$: constant phenotypic drift.

→ f degree 2 in x , V degree 0 :

$$D_f = 2 \quad D_V = 0$$

* Gaussian assumption from prior work

$$\hat{p}(t, x) \sim N(m_1(t), \sigma^2(t))$$

$$\text{where } \sigma^2(t) = m_2(t) - m_1^2(t)$$

$$p(t, x) = \frac{P(t)}{\sqrt{2\pi\sigma^2(t)}} \exp\left(-\frac{(x - m_1(t))^2}{2\sigma^2(t)}\right)$$

(If distribution is Gaussian, all higher moments are determined by mean & variance)

Prior work sub. Gaussian soln into PDE & matched coeff. to derive

ODE system

$$P' = \left[a - b(m_1 - X)^2 - b\sigma^2 \right] P - \frac{P^2}{K}$$

$$m_1' = -2b(m_1 - X)\sigma^2 + V_0$$

$$(\sigma^2)' = 2\beta - 2b\sigma^4$$

optimal
phenotype

Interpretation: $P' \rightarrow$ population grows fastest when m_1 close to X
 and distribution tightly conc. $-b\sigma^2$ penalizes pop if lots of variation.
 $-b(m_1 - X)$ penalizes if avg pheno is far from mean.
 • m_1' : mean pheno. changes for 2 reasons. No direct drift in phenotype space (environmental change pushing phenotypes in one direction).
 First term: $m_1 > X$ $m_1 - X > 0$, so term is negative and m_1 decreases towards X + opp. Can be said
 • σ^2 : diffusion ↑ phenotypic var., selection ↓ variance by favoring cells near optimal phenotype

The General moment framework can recover this system exactly, where before it could be derived substituting a Gaussian ansatz into PDE.
 For example, consider the P' equation,

$$\frac{dP}{dt} = \left(\int_{\mathbb{R}} f(t, x) \hat{p}(t, x) dx - \frac{P(t)}{K} \right) P(t)$$

$$\frac{dP}{dt} = \left(\int_{\mathbb{R}} [a(t) - b(t)(x - X(t))^2] \hat{p}(t, x) dx - \frac{P(t)}{K} \right) P(t)$$

$$\frac{dP}{dt} = \left(\int_{\mathbb{R}} a(t) \hat{p}(t, x) dx + \int_{\mathbb{R}} -b(t)(x^2 - 2xX(t) + X(t)^2) \hat{p} dx - \frac{P(t)}{K} \right) P(t)$$

$$= \left(a(t) + (-b(t)m_2(t) + 2b(t)X(t)m_1(t) - b(t)X(t)^2 - \frac{P(t)}{K}) \right) P(t)$$

Use $\sigma^2 = m_2 - m_1^2 \rightarrow m_2 = \sigma^2 + m_1^2$

$$= \left(a(t) - b(t)\sigma^2(t) - b(t)m_1^2(t) + 2b(t)X(t)m_1(t) - b(t)X(t)^2 - \frac{P(t)}{K} \right) P(t)$$

$$= \left(a(t) - b(t)\sigma^2(t) - b(t)(m_1(t) - X(t))^2 - \frac{P(t)}{K} \right) P(t)$$

So $P'(t) = \left[a(t) - b(t)(m_1(t) - X(t))^2 - b(t)\sigma^2(t) \right] P(t) - \frac{P^2(t)}{K}$
 which matches the above eqn.

Then, for $m_1'(t)$

$$m_1'(t) = \int_{\mathcal{R}} V \hat{p} dx + \int_{\mathcal{R}} x f \hat{p} dx - m_1 \int_{\mathcal{R}} f \hat{p} dx$$

$$m_1'(t) = \int_{\mathcal{R}} V_0(t) \hat{p} dx + \int_{\mathcal{R}} x [a - b(x - X)^2] \hat{p} dx - m_1 \int_{\mathcal{R}} [a - b(x - X)^2] \hat{p} dx$$

$$= V_0(t) + \int_{\mathcal{R}} [ax \hat{p} - b(x^3 - 2xXx^2 + xX^2) \hat{p}] dx - m_1 \int_{\mathcal{R}} [a \hat{p} - bx^2 \hat{p} + 2bXx \hat{p} - bX^2 \hat{p}] dx$$

$$= V_0(t) + a m_1(t) - b m_3(t) + 2bX m_2(t) - bX^2 m_1(t) - m_1 a + m_1 b m_2(t) - 2bX m_1^2 + m_1 b X^2$$

$$= V_0(t) - b m_3 + 2bX m_2 + b m_1 m_2 - 2bX m_1^2$$

$$= V_0(t) - b(m_1 m_2 + 2(m_2 - m_1^2) m_1) + 2bX m_2 + b m_1 m_2 - 2bX m_1^2$$

$$= V_0(t) - b m_1 m_2 - 2b(m_2 - m_1^2) m_1 + 2bX(m_2 - m_1^2) + b m_1 m_2$$

$$= V_0(t) - 2b m_1 \sigma^2 + 2bX \sigma^2 \quad \sigma^2 = m_2 - m_1^2$$

$$m_1'(t) = V_0(t) - 2b(t) (m_1(t) - X(t)) \sigma^2(t)$$

Which matches the one derived in earlier work.

We can do the same for $m_2'(t)$ to get an equation for $(\sigma^2)'(t)$. (since $\sigma^2 = \int (x - m_1)^2 \hat{p} dx = m_2 - m_1^2$)

Example shows point of paper: rather than

guessing Gaussian ansatz, authors derived
a general moment system that recovers
the Gaussian reduced model w/ one
closure choice. 